

✓ (0) Calcular las derivadas de las siguientes funciones:

a) $f(x) = (33 - 2x)^{\frac{4}{3}}$ d) $f(x) = \ln(7 - x)$ g) $f(x) = \ln(\cos(x) + \sin(x))$
 b) $f(x) = e^{2x}$ e) $f(x) = \ln(x^2 + 3x + 4)$ h) $f(x) = \frac{\cos(x)}{\sin(x)}$
 c) $f(x) = 2^x$ f) $f(x) = \ln(e^x + e^{-x})$

✓ (1) Dar las primitivas de las siguientes funciones:

a) $g(x) = x^3 - 5x$ c) $g(x) = \sin(2x)$ e) $g(x) = x^{3/2}$
 b) $g(x) = e^{0.3x}$ d) $g(x) = 2x \cos(x^2)$ f) $g(x) = \sqrt{x+2}$

0) a) $F(x) = (33 - 2x)^{\frac{4}{3}} = F'(x) = \frac{4}{3} (33 - 2x)^{\frac{1}{3}} (-2) = \left| \frac{-8 \sqrt[3]{33-2x}}{3} \right|$

b) $F(x) = e^{2x} = F'(x) = 2e^{2x}$

c) $F(x) = 2^x = e^{\ln(2)x} = \ln(2)x$

d) $F(x) = \ln(7 - x) = F'(x) = \frac{-1}{7-x}$

e) $F(x) = \left| \frac{2x+3}{x^2+3x+4} \right|$

f) $F(x) = \ln(e^x + e^{-x}) = F'(x) = \left| \frac{e^x - e^{-x}}{e^x + e^{-x}} \right|$

g) $F(x) = \ln(\cos(x) + \sin(x)) = F'(x) = \left| \frac{-\sin(x) + \cos(x)}{\cos(x) + \sin(x)} \right|$

h) $F(x) = \frac{\cos(x)}{\sin(x)} = F'(x) = \left| -\csc^2 \right|$

1) a) $g(x) = x^3 - 5x = \int x^3 dx - \int 5x dx = \left| \frac{x^4}{4} - \frac{5x^2}{2} + C \right|$

b) $g(x) = e^{0.3x}$

$= \int e^{kx} dx$ $\left| \begin{array}{l} k=0.3 \\ u=kx \\ du=k dx \end{array} \right|$ $\frac{1}{k} \int e^u du = \frac{1}{\frac{3}{10}} e^{kx} + C = \left| \frac{10}{3} e^{kx} + C \right|$

c) $g(x) = \sin(2x) = \int \sin(2x) dx = \left| -\frac{\cos(2x)}{2} + C \right|$

d) $g(x) = 2x \cos(x^2) = \int 2x \cos(x^2) dx$ $\left| \begin{array}{l} u=x^2 \\ du=2x du \end{array} \right| \int \cos(u) du$

$= \left| \sin(x^2) + C \right|$

e) $g(x) = x^{3/2} = \int x^{\frac{3}{2}} dx = \frac{x^{\frac{5}{2}}}{\frac{5}{2}} = \left| \frac{2x^{\frac{5}{2}}}{5} + C \right|$

f) $g(x) = \sqrt{x+2} = \int \sqrt{x+2} dx = \left| \begin{array}{l} u=x+2 \\ du=dx \end{array} \right| \int u^{\frac{1}{2}} dx = \left| \frac{2(x+2)^{\frac{3}{2}}}{\frac{3}{2}} + C \right|$

✓ (2) Encontrar la primitiva F de $f(x) = \frac{3}{x}$ tal que $F(1) = 5$.

(3) Calcular las siguientes integrales:

a) $\int e^{2x} dx$

d) $\int \frac{dx}{7-x}$

g) $\int \frac{\cos(x) - \sin(x)}{\cos(x) + \sin(x)} dx$

b) $\int 2^x dx$

e) $\int \frac{2x+3}{x^2+3x+4} dx$

h) $\int \frac{1}{\sin^2(x)} dx$

c) $\int \sqrt[3]{3-2x} dx$

f) $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$

Ayuda: usa el Ejercicio (0)

2) $F(x) = \frac{3}{x}$ $\rightarrow F(1) = 5$

$\int \frac{1}{x} dx = 3 \ln(|x|) + C \Rightarrow$ dado que, si $x=1 \Rightarrow 3 \ln(1) + C = 5$
 $\Rightarrow C = 5$
 $\Rightarrow \boxed{3 \ln(|x|) + 5}$

3) a) $\int e^{2x} dx = \left| \begin{matrix} u=2x \\ du=2du \end{matrix} \right| \frac{1}{2} \int e^u du = \left| \frac{e^u}{2} + C \right|$

b) $\int 2^x dx = \left| \frac{2^x}{\ln(2)} + C \right|$

c) $\int \sqrt[3]{3-2x} dx = \left| \begin{matrix} u=3-2x \\ du=-2du \end{matrix} \right| = -\frac{1}{2} \int u^{\frac{1}{3}} du = -\frac{1}{2} \frac{u^{\frac{4}{3}}}{\frac{4}{3}} = \left| -\frac{3}{8} (3-2x)^{\frac{4}{3}} + C \right|$

d) $\int \frac{1}{7-x} dx = \left| -\ln(|7-x|) + C \right|$

e) $\int \frac{2x+3}{x^2+3x+4} dx = \left| \ln(|x^2+3x+4|) + C \right|$

f) $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \left| \ln(|e^x + e^{-x}|) + C \right|$

g) $\int \frac{\cos(x) - \sin(x)}{\cos(x) + \sin(x)} dx = \left| \ln(|\cos(x) + \sin(x)|) + C \right|$

h) $\int \frac{1}{\sin^2(x)} dx = \int \csc^2(x) dx = \left| -\cotan(x) + C \right|$

✓ (4) Sin realizar el cálculo de la integral, justificar las siguientes igualdades y desigualdades:

a) $\int_{-\pi}^{\pi} \sin(2x) dx = 0$

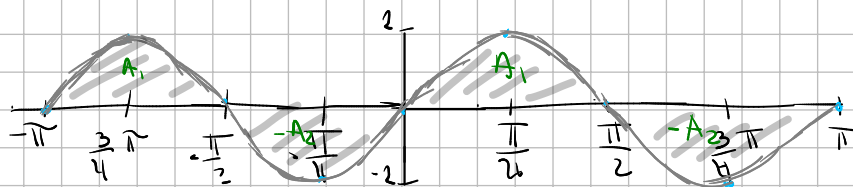
b) $\pi/6 \leq \int_{\pi/6}^{\pi/2} \sin(x) dx \leq \pi/3$

c) $\int_1^2 \sqrt{5-x} dx \geq \int_1^2 \sqrt{x+1} dx$

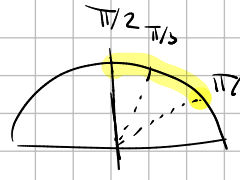
✓ (5) Calcular la derivada de las siguientes funciones donde sea posible:

a) $f(x) = \int_0^x \frac{\sin(t^2)}{1+\cos^2 t} dt$ b) $f(x) = \int_0^{x^2} \frac{e^{t^2}+1}{\sqrt{1-t^2}} dt$ c) $f(x) = \int_{\sqrt{x}}^{x^3} \frac{t+1}{\sqrt{1+2t}} dt$

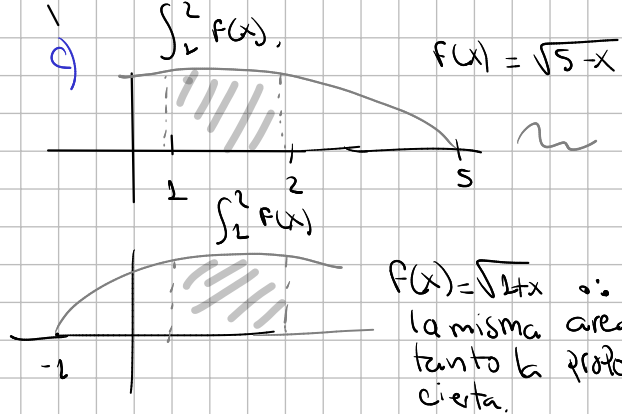
1) a)



Dado que las áreas son las mismas, se restan.

b)  $\therefore$ Dado que $\pi/6 \leq \pi/3$ la desigualdad es cierta.

Sa)
$$f(x) = \int_0^x \frac{\sin(t^2)}{1+\cos^2 t} dt$$



$f'(x) = g(b(x)) \cdot b'(x) - g(a(x)) \cdot a'(x)$

utilizo la regla de Leibniz

$$f'(x) = \frac{\sin(x^2)}{1+\cos^2(x)}$$

b)
$$f(x) = \int_0^{x^2} \frac{e^{t^2} + 1}{\sqrt{1-t^2}} dt \Rightarrow f'(x) = \frac{e^{x^4} + 1 \cdot 2x}{\sqrt{1-x^4}}$$

c)
$$f(x) = \int_{\sqrt{x}}^{x^3} \frac{t+1}{\sqrt{1+t^2}} dt \Rightarrow f'(x) = \frac{x^3+1}{\sqrt{1+x^6}} \cdot 3x^2 - \frac{\sqrt{x}+1}{\sqrt{1+2\sqrt{x}}} \cdot \frac{1}{2\sqrt{x}}$$

✓ (6) Calcular las siguientes integrales usando el Teorema Fundamental del Cálculo:

a) $\int_1^2 2^x dx$

c) $\int_1^5 \frac{dx}{7-x}$

e) $\int_{\ln 2}^{\ln 3} \frac{e^x - e^{-x}}{e^x + e^{-x}} dx$

b) $\int_3^5 \sqrt[3]{33-2x} dx$

d) $\int_0^1 \frac{2x+3}{x^2+3x+4} dx$

f) $\int_0^{\pi/2} \frac{\cos(x) - \sin(x)}{\cos(x) + \sin(x)} dx$

a)
$$\int_1^2 2^x dx = \left. \frac{2^x}{\ln(2)} \right|_1^2 = \frac{11}{\ln(2)} - \frac{2}{\ln(2)} = \boxed{\frac{9}{\ln(2)}}$$

b)
$$\int_3^5 \sqrt[3]{33-2x} dx = \left| \begin{array}{l} u = 33-2x \\ du = -2 du \end{array} \right| - \frac{1}{2} \int_{u(3)}^{u(5)} u^{\frac{1}{2}} du = -\frac{u^{\frac{3}{2}}}{\frac{3}{2}} \Big|_{u(3)}^{u(5)}$$

$$= -\frac{(33-2x)^{\frac{3}{2}}}{\frac{3}{2}} \Big|_3^5 = -\frac{23^{\frac{3}{2}}}{\frac{3}{2}} + \frac{27^{\frac{3}{2}}}{\frac{3}{2}} = \boxed{\frac{27^{\frac{3}{2}} - 23^{\frac{3}{2}}}{\frac{3}{2}}}$$

c)
$$\int_1^5 \frac{1}{7-x} dx = -\ln(|7-x|) \Big|_1^5 = -\ln(12) + \ln(16) = \ln\left(\frac{6}{2}\right) = \boxed{\ln(3)}$$

d)
$$\int_0^1 \frac{2x+3}{x^2+3x+4} dx = \ln(|x^2+3x+4|) \Big|_0^1 = \ln(8) - \ln(4) = \boxed{\ln(2)}$$

e)
$$\int_{\ln(2)}^{\ln(3)} \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \ln(|e^x + e^{-x}|) \Big|_{\ln(2)}^{\ln(3)} = \ln(13-2) - \ln(12-2) = \boxed{-1}$$

f) $\int_0^{\pi/2} \frac{\cos(x) - \sin(x)}{\cos(x) + \sin(x)} dx = \ln(|\cos(x) + \sin(x)|) \Big|_0^{\pi/2} = \boxed{0}$

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(7) Calcular las siguientes integrales:

a) $\int x e^x dx$

d) $\int_{\pi/4}^{\pi/2} \frac{x dx}{\sin^2(x)}$

g) $\int_0^2 x \ln(x^2 + 4) dx$

b) $\int_{-1}^1 (1-2x) e^{-2x} dx$

e) $\int_3^9 x \ln(x-1) dx$

h) $\int e^{-x} \sin(2x) dx$

c) $\int x^2 \cos(x) dx$

f) $\int_3^9 \ln(x^2 + 1) dx$

i) $\int_0^{2\pi} \cos^4(x) dx$

a) $\int x e^x dx \quad \left| \begin{array}{l} u = x \\ du = dx \end{array} \quad \begin{array}{l} dv = e^x dx \\ v = e^x \end{array} \right|$

$x e^x - \int e^x dx = x e^x - e^x + C = \underline{e^x(x-1) + C}$

b) $\int_{-1}^1 (1-2x) e^{-2x} dx = \left| \begin{array}{l} u = (1-2x) \\ du = -2 dx \end{array} \quad \begin{array}{l} dv = e^{-2x} \\ v = -\frac{e^{-2x}}{2} \end{array} \right|$

$-\frac{(1-2x)e^{-2x}}{2} - \int \frac{e^{-2x}}{2} (-2) dx = -\frac{(1-2x)e^{-2x}}{2} + \frac{e^{-2x}}{2} \Big|_{-1}^1 = \frac{e^{-2x}((2x-1)+1)}{2} \Big|_{-1}^1$

$\frac{e^{-2x} \cdot 2x}{2} \Big|_{-1}^1 = \underline{e^{-2} + e^2}$

c) $\int x^2 \cos(x) dx = \left| \begin{array}{l} u = x^2 \\ du = 2x dx \end{array} \quad \begin{array}{l} dv = \cos(x) dx \\ v = \sin(x) \end{array} \right| \quad \left| \begin{array}{l} u = 2x \\ du = 2 dx \end{array} \quad \begin{array}{l} dv = \sin(x) \\ v = -\cos(x) \end{array} \right|$

$x^2 \sin(x) - \int \sin(x) 2x dx = x^2 \sin(x) - \left[2x \cos(x) + \frac{1}{2} \int \cos(x) dx \right]$

$= \underline{x^2 \sin(x) + 2x \cos(x) - 2 \sin(x) + C}$

d) $\int_{\pi/4}^{\pi/2} \frac{x}{\sin^2(x)} dx \quad \left| \begin{array}{l} u = x \\ du = dx \end{array} \quad \begin{array}{l} dv = \frac{1}{\sin^2(x)} \\ v = -\frac{\cos(x)}{\sin(x)} \end{array} \right| \quad \left| \begin{array}{l} u = \sin(x) \\ du = \cos(x) dx \end{array} \right|$

$-\frac{x \cos(x)}{\sin(x)} + \int \frac{\cos(x)}{\sin(x)} dx = -x \frac{\cos(x)}{\sin(x)} - \ln(|\sin(x)|) \Big|_{\pi/4}^{\pi/2} = \underline{\frac{\pi}{4} + \ln\left(\frac{\sqrt{2}}{2}\right)}$

e) $\int_3^9 x \ln(x-2) dx = \frac{\ln(x-1)}{2} x^2 - \frac{1}{2} \int \frac{x^2}{x-1} dx = \left| \begin{array}{l} u = \ln(x-2) \\ du = \frac{1}{x-2} dx \end{array} \quad \begin{array}{l} dv = x \\ v = \frac{x^2}{2} \end{array} \right|$

$\left| \begin{array}{l} u = x-1 \\ du = dx \end{array} \right|$

$$\frac{\ln(x-1)x^2}{2} - \frac{1}{2} \int \frac{(u+1)^2}{u} = \ln \frac{(x-1)x^2}{2} - \frac{u^2}{2} - \ln(u) - u =$$

$$\ln \frac{(x-1)x^2}{2} - \frac{(x-1)^2}{4} - \ln(x-1) - x-1 \Big|_3^9 = \left(\frac{\ln(8)81}{2} - 26 - \ln(181) \right) - \left(\frac{\ln(13)9}{2} - 5 - \ln(121) \right)$$

$$\int \ln(x^2+1) dx = \ln(x^2+1)x - 2 \int \frac{x^2}{x^2+1} dx =$$

$$\ln(x^2+1)x - 2 \int \frac{x^2+1}{x^2+1} dx + 2 \int \frac{1}{x^2+1} dx$$

$$\ln(x^2+1)x - 2x + 2 \arctan(x) + C$$

$$\left| \begin{array}{ll} u = \ln(x^2+1) & dv = dx \\ du = \frac{2x}{x^2+1} dx & v = x \end{array} \right|$$

$$\left| \begin{array}{ll} u = x^2 & dv = \frac{1}{x^2+1} \\ du = 2x dx & v = \arctan(x) \end{array} \right|$$

$$g) \int_0^2 x \ln(x^2+4) dx \quad \left| \begin{array}{l} u = x^2+4 \\ du = 2x dx \end{array} \right| \quad \frac{1}{2} \int_{u(0)}^{u(2)} \ln(u) du$$

$$\frac{1}{2} \left[\ln(u) \cdot u - \int \frac{u}{u} du \right]_{u(0)}^{u(2)} = \frac{(x^2+4)(\ln(x^2+4) - 1)}{2} \Big|_0^2 = \frac{(8 \ln(8) - 8)}{2} - \frac{(4 \ln(4) - 4)}{2}$$

$$= \frac{8 \ln(8) - 4 \ln(4) - 4}{2}$$

$$h) \int e^{-x} \sin(2x) dx$$

$$i) \int_0^{2\pi} \cos^4(x) dx =$$

9. Trazar la región limitada por las curvas dadas y calcular su área:

a) $y = 4x^2$, $y = x^2 + 3$

b) $y = \cos(x)$, $y = \sin(x)$, $x = 0$, $x = \pi/2$.

c) $y = |x|$, $y = (x+1)^2 - 7$, $x = -4$

a) $4x^2 = x^2 + 3$

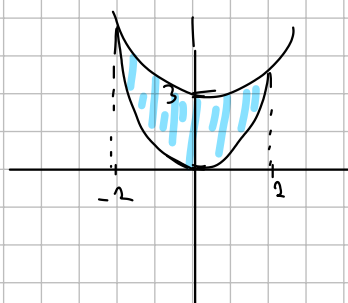
$$3x^2 = 3$$

$$x = \pm\sqrt{1} \Rightarrow \boxed{x_1 = -1} \vee \boxed{x_2 = 1}$$

$$\Rightarrow \left. \begin{aligned} -x^3 \Big|_{-1}^1 &= -1 - (-2) = 1 \\ 3x \Big|_{-1}^1 &= 3 + 3 = 6 \end{aligned} \right\} \Rightarrow \boxed{A = 4}$$

$$\int_{-1}^1 ((x^2 + 3) - 4x^2) dx$$

$$= \int_{-1}^1 (-3x^2 + 3) dx = -3 \int_{-1}^1 x^2 dx + \int_{-1}^1 3 dx = -\frac{3x^3}{3} + 3x \Big|_{-1}^1$$

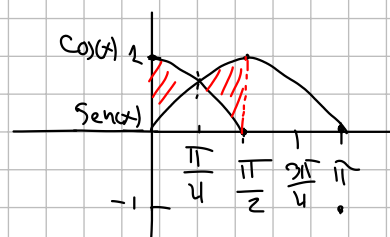


b) $\sin(x) = \cos(x)$ en $\pi/4$

$$\Rightarrow \int_0^{\pi/4} (\cos(x) - \sin(x)) dx + \int_{\pi/4}^{\pi/2} (\sin(x) - \cos(x)) dx$$

$$\Rightarrow \sin(x) + \cos(x) \Big|_0^{\pi/4} = \boxed{\sqrt{2} - 1}$$

$$\Rightarrow -\cos(x) - \sin(x) \Big|_{\pi/4}^{\pi/2} = -1 - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right) = \boxed{-1 + \sqrt{2}} \Rightarrow A = \sqrt{2} - 1 + \sqrt{2} - 1 = \boxed{2\sqrt{2} - 2}$$

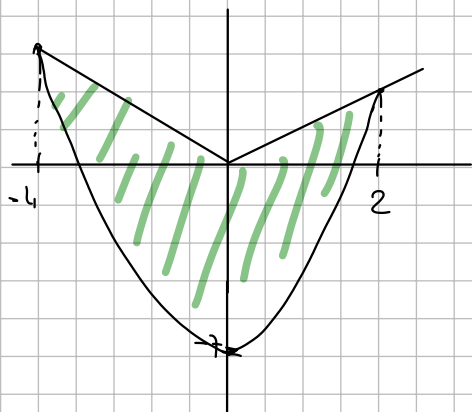


c) $x > 0 \mid x = (x+1)^2 - 7$
 $0 = x^2 + x - 6$

$-1 \pm \sqrt{1+24} = \boxed{x_1 = 2}$
 $\boxed{x_2 = -3} \rightarrow X \text{ es } < 0, \text{ no vale}$

$x < 0 \mid 0 = x^2 + 3x - 6$
 $-3 \pm \sqrt{33} =$

Solo es válida la negativa



8. Calcular las siguientes integrales:

-a) $\int_0^1 e^{\sqrt{x}} dx$ -d) $\int \frac{1}{x \ln(x)} dx$ -g) $\int e^x (1 - e^x)^{-1} dx$
 -b) $\int \sin(\sqrt{x}) dx$ -e) $\int_0^1 \arccos(x) dx$ ✓h) $\int \frac{dx}{x\sqrt{x-1}}$
 -c) $\int_0^1 (2x+1) \ln(x+1) dx$ -f) $\int_0^1 x^3 e^{x^2} dx$ ✓i) $\int \sin^3(x) dx$

a) $\int_0^1 e^{\sqrt{x}} dx = \left| \begin{array}{l} u = \sqrt{x} \\ du = \frac{1}{2\sqrt{x}} dx \end{array} \right| 2 \int_{u(0)}^{u(1)} u e^u du \quad \left| \begin{array}{l} u = u \\ du = dx \end{array} \right| \quad \left| \begin{array}{l} dv = e^u dx \\ v = e^u \end{array} \right|$

$u e^u - \int e^u du = 2\sqrt{x} e^{\sqrt{x}} - 2e^{\sqrt{x}} \Big|_0^1 = -(-2) = \boxed{2} //$

b) $\int \sin(\sqrt{x}) dx \quad \left| \begin{array}{l} u = \sqrt{x} \\ du = \frac{1}{2\sqrt{x}} dx \end{array} \right| 2 \int u \sin(u) du = \left| \begin{array}{l} u = u \\ du = dx \\ dv = \sin(u) \\ v = -\cos(u) \end{array} \right|$

$2 \left[-u \cos(u) + \int \cos(u) du \right] = \boxed{-2\sqrt{x} \cos(\sqrt{x}) + 2 \sin(\sqrt{x}) + C} //$

c) $\int_0^1 (2x+1) \ln(x+1) dx = \left| \begin{array}{l} u = \ln(x+1) \\ du = \frac{1}{x+1} dx \end{array} \right| \quad \left| \begin{array}{l} dv = (2x+1) \\ v = x^2+x \end{array} \right|$

$\ln(x+1)(x^2+x) - \int \frac{x(x+1)}{x+1} dx = \ln(x+1)(x^2+x) - \frac{x^2}{2} \Big|_0^1 = \boxed{2 \ln(2) - \frac{1}{2}}$

$$d) \int \frac{1}{x \ln(x)} dx = \left| \begin{array}{l} u = \ln(x) \\ du = \frac{1}{x} dx \end{array} \right| \int \frac{1}{u} du = \boxed{\ln(|\ln(x)|) + C}$$

$$e) \int_0^1 \arccos(x) dx = x \arccos(x) - \sqrt{1-x^2} \Big|_0^1 = \boxed{\frac{1}{2}}$$

$$f) \int_0^1 x^3 e^{x^2} dx \quad \left| \begin{array}{l} u = x^2 \\ du = 2x dx \\ \frac{du}{2x} = dx \end{array} \right| \frac{1}{2} \int_{u(0)}^{u(1)} u e^u du = \left| \begin{array}{l} u = u \\ du = dx \\ v = e^u \end{array} \right|$$

$$\frac{1}{2} [u e^u - \int e^u du] = \frac{x^2 e^{x^2}}{2} - \frac{e^{x^2}}{2} \Big|_0^1 = \frac{e^{x^2}(x^2 - 1)}{2} \Big|_0^1 = \boxed{-\frac{e}{2}}$$

$$g) \int \frac{e^x}{1-e^x} dx \quad \left| \begin{array}{l} u = e^x \\ du = e^x dx \end{array} \right| \int \frac{1}{1-u} du \quad \left| \begin{array}{l} u = 1-u \\ du = -dx \end{array} \right|$$

$$-\int \frac{1}{u} du = \boxed{-\ln(|1-e^x|) + C}$$

$$i) \int \sin^3(x) dx = \int (1 - \cos^2(x)) \sin(x) dx \quad \left| \begin{array}{l} u = \cos(x) \\ du = -\sin(x) dx \end{array} \right| \int (1-u^2) du$$

$$= \int 1 du - \int u^2 du = u - \frac{u^3}{3} = \boxed{\cos(x) - \frac{\cos^3(x)}{3}}$$

$$h) \int \frac{1}{x\sqrt{x-1}} dx \quad \left| \begin{array}{l} u = \sqrt{x-1} \\ du = \frac{1}{2\sqrt{x-1}} \\ x = u^2 + 1 \end{array} \right| 2 \int \frac{1}{u^2+1} du = \boxed{2 \arctan(\sqrt{x-1}) + C}$$

10. Calcular las siguientes integrales:

$$a) \int_2^4 \frac{x^2 + 4x + 24}{x^2 - 4x + 8} dx$$

$$b) \int_0^2 \frac{x-1}{x^2+4} dx$$

$$c) \int_2^4 \frac{x}{x^3-3x+2} dx$$

$$b) \int_0^2 \frac{x-1}{x^2+4} dx = \int_0^2 \frac{x}{x^2+4} dx - \int_0^2 \frac{1}{x^2+4} dx = \left| \begin{array}{l} u = x^2+4 \\ du = 2x dx \end{array} \right|$$

$$\Rightarrow \frac{\ln(|x^2+4|)}{2} - \frac{\arctan(\frac{x}{2})}{2} \Big|_0^2 = \left(\frac{\ln(8)}{2} - \frac{\arctan(2)}{2} - \frac{\ln(4)}{2} \right) = \boxed{\frac{\ln(2)}{2} - \frac{\arctan(2)}{2}}$$

$$c) \int_2^4 \frac{x}{x^3-3x+2} dx = \int_2^4 \frac{x}{(x-1)^2(x+2)} dx = \int_2^4 \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2} dx =$$

$$\frac{A(x-1)(x+2) + B(x+2) + C(x-1)^2}{(x-1)^2(x+2)} = \int_2^4 \frac{(A+C)x^2 + (A+B-2C)x + (-2A+2B+C)}{(x-1)^2(x+2)} dx =$$

$$\Rightarrow \frac{2}{9} \int_2^4 \frac{1}{(x-1)} dx + \frac{1}{3} \int_2^4 \frac{1}{(x-1)^2} dx - \frac{2}{9} \int_2^4 \frac{1}{(x+2)} dx \quad \left\{ \begin{array}{l} \bullet A+C=0 \Rightarrow A=-C \\ \bullet A+B-2C=1 \text{ (1)} \\ \bullet -2A+2B+C=0 \text{ (2)} \end{array} \right. \quad \left\{ \begin{array}{l} A = \frac{2}{9} \\ C = -\frac{2}{9} \\ B = \frac{1}{3} \end{array} \right.$$

$$\left. \frac{2}{9} \ln(1x-11) + \frac{1}{3} \ln(1x-11) - \frac{2}{9} \ln(1x+21) \right|_2^4$$

$$= \left(\frac{2}{9} \ln(131) + \frac{1}{3} \ln(131) - \frac{2}{9} \ln(121) \right) + \frac{2}{9} \ln(141) = \left[\frac{2}{9} \left(\ln(131) + \ln(141) - \ln(121) \right) + \frac{1}{3} \ln(131) \right]$$

a) $\int_2^4 \frac{x^2 + 4x + 21}{x^2 - 4x + 8} dx =$

$$\begin{array}{r} x^2 + 4x + 21 \quad | \quad x^2 - 4x + 8 \\ -x^2 + 4x - 8 \\ \hline 8x + 16 \\ \quad \quad \quad \swarrow (8x + 16) \\ \quad \quad \quad 0 \end{array}$$

$$\Rightarrow \int_2^4 \frac{8x + 16}{(x-2)^2 + 4} dx + \int_2^4 1 dx =$$

$$\textcircled{1} \int_2^4 \frac{8(\cancel{x-2}) + 32}{(x-2)^2 + 4} dx = 8 \int_2^4 \frac{1}{x+2} dx + 32 \int_2^4 \frac{1}{((x-2)^2 + 2^2)} dx$$

$$\Rightarrow 8 \ln(1x+21) + 16 \arctg\left(\frac{x-2}{2}\right) + x \Big|_2^4 = (8 \ln(161) + 16 \arctg(1) + 4) - (8 \ln(141) + 2)$$

$$= \left[8 \ln(161) + 4\pi - 8 \ln(141) + 2 \right]$$

11. La sustitución $t = \tan\left(\frac{x}{2}\right)$, o equivalentemente, $x = 2 \arctan(t)$, transforma cualquier integral que involucre sólo senos y cosenos vinculados por suma, producto o cociente, en la integral de una función racional. Verificar que con esta sustitución resulta

$$\cos(x) = \frac{1-t^2}{1+t^2}, \quad \sin(x) = \frac{2t}{1+t^2} \quad y \quad dx = \frac{2}{1+t^2} dt.$$

Utilizar esta sustitución en los siguientes casos:

a) $\int_0^{\pi/2} \frac{2}{1+\cos(x)} dx$ b) $\int_{\pi/3}^{\pi/2} \frac{1}{\sin(x)} dx$

a) $\int_0^{\pi/2} \frac{2}{1+\cos(x)} dx = 2 \int_0^{\pi/2} \frac{1}{1 + \frac{1-t^2}{1+t^2}} dx = \int_0^{\pi/2} \frac{1+t^2 + 1-t^2}{1+t^2} dx = \int_0^{\pi/2} \frac{2}{1+t^2} dx = \frac{2}{2+t^2} \Big|_0^{\pi/2} = \frac{2}{2+t^2} \Big|_0^{\pi/2}$

12. Calcular las siguientes integrales:

a) $\int \tan^2(x) dx$

c) $\int \frac{dx}{\sqrt{9-4x^2}}$

e) $\int \frac{x+1}{\sqrt{1-x^2}} dx$

a) $\int \tan^2(x) dx = \int (\sec^2 - 1) dx = \int \sec^2(x) dx - x + C = \boxed{\tan(x) - x + C}$

b) $\int \frac{dx}{\sqrt{9-4x^2}} = \frac{1}{2} \int \frac{1}{\sqrt{3^2 - (2x)^2}} du = \left| \begin{array}{l} u = 2x \\ du = 2 dx \end{array} \right| \boxed{\frac{1}{2} \arcsin\left(\frac{2x}{3}\right) + C}$

e) $\int \frac{x+1}{\sqrt{1-x^2}} dx = \int \frac{x}{\sqrt{1-x^2}} dx + \arcsin(x) = -\frac{1}{2} u^{-1/2} + \arcsin(x) = \left| \begin{array}{l} u = 1-x^2 \\ du = -2x dx \end{array} \right| \boxed{-\frac{1}{\sqrt{1-x^2}} + \arcsin(x) + C}$

13. Determinar si las siguientes integrales impropias convergen y en tal caso calcularlas.

a) $\int_0^{+\infty} \frac{1}{\sqrt{s+1}} ds$

b) $\int_0^2 \frac{1}{(1-y)^{2/3}} dy$

c) $\int_{-\infty}^0 x e^{-x^2} dx$

a) $\int_0^{\infty} \frac{1}{\sqrt{s+1}} ds = \lim_{t \rightarrow \infty} \int_0^t \frac{1}{\sqrt{s+1}} ds \left| \begin{array}{l} u = s+1 \\ du = ds \end{array} \right| \lim_{t \rightarrow \infty} \int_{u(0)}^{u(t)} u^{-1/2} du = \lim_{t \rightarrow \infty} 2\sqrt{s+1} \Big|_0^t$

$\lim_{t \rightarrow \infty} (2\sqrt{t+1} - 2) = \infty \therefore \text{diverge}$

c) $\int_{-\infty}^0 x e^{-x^2} dx = \lim_{t \rightarrow -\infty} \int_t^0 x e^{-x^2} dx \left| \begin{array}{l} u = -x^2 \\ du = -2x dx \end{array} \right| -\frac{1}{2} \int e^u du = \lim_{t \rightarrow -\infty} -\frac{1}{2} e^{-x^2} \Big|_t^0$

$\lim_{t \rightarrow -\infty} \left(-\frac{1}{2} + \frac{1}{2e^{t^2}} \right) = \boxed{-\frac{1}{2}} \therefore \text{Converge}$

$\left| \begin{array}{l} u = 1-y \\ du = -dy \end{array} \right|$

b) $\int_0^2 \frac{1}{(1-y)^{2/3}} dy = \lim_{t \rightarrow -1} - \int_0^t u^{-2/3} = \lim_{t \rightarrow 1^-} \int_t^2 (1-y)^{-2/3} dy = \lim_{t \rightarrow 1^-} \left(-3\sqrt[3]{1-y} \Big|_0^t \right) + \lim_{t \rightarrow 1^+} \left(-3\sqrt[3]{1-y} \Big|_t^2 \right)$
 $= \lim_{t \rightarrow 1^-} (-3\sqrt[3]{1-t} + 3) + \lim_{t \rightarrow 1^+} (-3\sqrt[3]{1-2} + 3\sqrt[3]{1-t}) = \boxed{16} \therefore \text{Converge}$

14. Determinar si cada una de las siguientes integrales impropias converge o no.

a) $\int_4^{+\infty} \frac{1}{\sqrt{s}-1} ds$

c) $\int_0^4 \frac{dx}{(x-3)^{2/3}}$

e) $\int_0^4 \frac{dx}{x^2-x-2}$

a) $\int_4^{\infty} \frac{1}{\sqrt{s}-1} ds = \left| \begin{array}{l} u = \sqrt{s}-1 \\ du = \frac{1}{2\sqrt{s}} \end{array} \right|$
 $\lim_{t \rightarrow \infty} 2 \int_4^t \frac{u+1}{u} du = \lim_{t \rightarrow \infty} \left(\int_{u1}^t 1 du + \int_{u1}^t \frac{1}{u} du \right) = \lim_{t \rightarrow \infty} \left[(\sqrt{s}-1) \Big|_4^t + \ln(\sqrt{s}-1) \Big|_4^t \right]$

$$= \lim_{t \rightarrow \infty} \left[(\sqrt{t}-1) + \ln(\sqrt{t}-1) \right] = \lim_{t \rightarrow \infty} 2\sqrt{t} + 2\ln(\sqrt{t}-1) = \infty \therefore \text{diverge}$$

$$b) \int_0^4 \frac{1}{(x-3)^{2/3}} dx = \lim_{t \rightarrow 3^-} \int_0^t (x-3)^{-2/3} dx + \lim_{t \rightarrow 3^+} \int_t^4 (x-3)^{-2/3} dx \quad \left| \begin{array}{l} u = x-3 \\ du = dx \end{array} \right|$$

$$= \lim_{t \rightarrow 3^-} 3u^{1/3} \Big|_0^t + \lim_{t \rightarrow 3^+} 3u^{1/3} \Big|_t^4 = \lim_{t \rightarrow 3^-} 3\sqrt[3]{x-3} \Big|_0^t + \lim_{t \rightarrow 3^+} 3\sqrt[3]{x-3} \Big|_t^4$$

$$= \lim_{t \rightarrow 3^-} (3\sqrt[3]{t-3} - 3\sqrt[3]{-3}) + \lim_{t \rightarrow 3^+} (3 - 3\sqrt[3]{t-3}) = \boxed{-3\sqrt[3]{-3} + 3} \therefore \text{Converge}$$

$$c) \int_0^4 \frac{1}{x^2-x-2} dx$$

$$= \int_0^4 \frac{1}{(x+1)(x-2)} dx = \frac{A(x-2) + B(x+1)}{(x+1)(x-2)} = \frac{(A+B)x + (B-2A)}{(x+1)(x-2)} =$$

$$\Rightarrow \begin{cases} A+B=0 \\ B-2A=1 \end{cases} \quad \begin{array}{l} \textcircled{1} A = -B = \boxed{-\frac{1}{3}} \\ \textcircled{2} B + 2B = 1 \\ \boxed{B = \frac{1}{3}} \end{array} \left\{ \begin{array}{l} -\frac{1}{3} \int_0^4 \frac{1}{(x+1)} dx + \frac{1}{3} \int_0^4 \frac{1}{x-2} dx \\ = -\frac{1}{3} \ln(|x+1|) + \frac{1}{3} \ln(|x-2|) \Big|_0^4 \end{array} \right.$$

$$= \left(-\frac{1}{3} \ln(5) + \frac{1}{3} \ln(2) \right) - \left(-\frac{1}{3} \ln(1) + \frac{1}{3} \ln(-2) \right) = \boxed{-\frac{1}{3} \ln(5)} \therefore \text{diverge, estoy analizando un área } \oplus$$

y obtengo un valor $\ominus$

Ejercicios adicionales

6. Calcular las siguientes integrales:

✓ a) $\int_0^1 \frac{2x+1}{x^2+1} dx$ ✓ b) $\int_2^3 \frac{1}{x^2+3x+2} dx$ ✓ c) $\int \frac{x^3}{(x^2+1)^3} dx$

Ayuda: en el inciso (c) sustituya $u = x^2 + 1$.

$u = x^2 + 1$
 $du = 2x dx$

a) $\int_0^1 \frac{2x+1}{x^2+1} dx = 2 \int_0^1 \frac{x dx}{x^2+1} + \int_0^1 \frac{1}{x^2+1} dx = \arctan(x) + \ln(|x^2+1|) \Big|_0^1$
 $= \left(\arctan(1) + \ln(2) \right) - \arctan(0) = \left[\frac{\pi}{4} + \ln(2) \right]$

b) $\int_2^3 \frac{1}{x^2+3x+2} dx$
 $= \int_2^3 \frac{1}{(x+1)(x+2)} dx = \int_2^3 \left(\frac{A}{x+1} + \frac{B}{x+2} \right) dx = \frac{A(x+2) + B(x+1)}{(x+1)(x+2)} = \frac{x(A+B) + (2A+B)}{(x+1)(x+2)}$

$\begin{cases} A+B=0 & (1) & (1) & A=-B \Rightarrow \boxed{A=1} \\ 2A+B=1 & (2) & (2) & -B=1 \end{cases} \Rightarrow \boxed{B=-1}$
 $\Rightarrow \int_2^3 \left(\frac{1}{x+1} - \frac{1}{x+2} \right) dx = \int_2^3 \frac{1}{x+1} dx - \int_2^3 \frac{1}{x+2} dx$
 $= \ln(|x+1|) - \ln(|x+2|) \Big|_2^3 = (\ln(4) - \ln(5)) - (\ln(3) - \ln(4))$
 $= \ln(4) - \ln(5) - \ln(3) + \ln(4) = 2\ln(4) - \ln(3) - \ln(5)$

c) $\int \frac{x^3}{(x^2+1)^3} dx = \frac{1}{2} \int \frac{(u-1)}{u^3} du = \frac{1}{2} \int \frac{u-1}{u^3} du$
 $= \frac{1}{2} \int u^{-2} du - \frac{1}{2} \int u^{-3} du = \left[-\frac{(x^2+1)^{-1}}{2} + \frac{(x^2+1)^{-2}}{4} \right] + C$
 $\int \frac{u}{u^3} = \int \frac{1}{u^2} = -\frac{1}{u}$

7. Calcular las siguientes integrales:

b) $\int_4^9 \frac{\sqrt{x}+1}{\sqrt{x}-1} dx$ d) $\int_1^3 \frac{dx}{\sqrt{12x-8-3x^2}}$ f) $\int_0^{100\pi} \sqrt{1-\cos(2x)} dx$

b) $\int_4^9 \frac{\sqrt{x}+1}{\sqrt{x}-1} dx =$
 $u = \sqrt{x} - 1$
 $du = \frac{1}{2\sqrt{x}}$

$$d) \int_1^3 \frac{1}{\sqrt{-3x^2 + 12x - 9}} dx =$$

$$\int_1^3 \frac{1}{\sqrt{-3(x-2)^2 + 2^2}} dx = \int_1^3 \frac{1}{\sqrt{-\left(\sqrt{3}(x-2)\right)^2 + 2^2}} dx$$

$$\begin{aligned} \sqrt{3}x - \\ u = \sqrt{3}(x-2) \\ du = \sqrt{3} dx \end{aligned}$$

$$\frac{1}{\sqrt{3}} \int_1^3 \frac{1}{\sqrt{-u^2 + 2^2}} du = \frac{\arcsen\left(\frac{\sqrt{3}(x-2)}{2}\right)}{\sqrt{3}} \Big|_1^3 = \frac{\arcsen\left(\frac{\sqrt{3}}{2}\right)}{\sqrt{3}} - \frac{\arcsen\left(\frac{-\sqrt{3}}{2}\right)}{\sqrt{3}}$$

$$F) \int_0^{100\pi} (1 - \cos^2(2x)) dx = \int_0^{100\pi} |\sin(2x)| dx =$$

$$\begin{aligned} u = 2x \\ du = 2 dx \end{aligned}$$

$$= \frac{1}{2} \int_0^{100\pi} \sin(u) du = -\frac{1}{2} \cos(2x) \Big|_0^{100\pi} = -\frac{1}{2} + \frac{1}{2} = \boxed{0} //$$

✓ 8. Determinar si las siguientes integrales impropias convergen y en tal caso calcularlas.

✓ d) $\int_{-1}^7 \frac{dx}{\sqrt[3]{x+1}}$

✓ e) $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$

✓ f) $\int_0^1 \ln(x) dx$

9. Determinar si cada una de las siguientes integrales impropias converge o no.

✓ b) $\int_0^{\infty} e^{-x} \cos(x) dx$

✓ d) $\int_0^1 x \ln(x) dx$

✓ f) $\int_1^{\infty} \frac{\sin(x)}{x^2} dx$

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d) $\int_{-1}^7 \frac{1}{\sqrt[3]{x+1}} dx = \lim_{t \rightarrow -1} \int_t^7 \frac{1}{\sqrt[3]{x+1}} dx = \left| \begin{matrix} u = x+1 \\ du = dx \end{matrix} \right| = \lim_{t \rightarrow -1} \int_t^7 u^{-1/3} du = \lim_{t \rightarrow -1} \left. \frac{3}{2} (x+1)^{2/3} \right|_t^7$
 $= \lim_{t \rightarrow -1} \left(\frac{3 \cdot 2^{2/3}}{2} - \frac{3(t+1)^{2/3}}{2} \right) = \frac{3 \cdot 4}{2} - \lim_{t \rightarrow -1^+} \frac{3 \sqrt[3]{(t+1)^2}}{2} = \boxed{6}$

∴ la integral converge y su valor es 6.

e) $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \int_{-\infty}^0 \frac{dx}{1+x^2} + \int_0^{\infty} \frac{dx}{1+x^2} =$ ∴ Por lo tanto converge a $\boxed{\pi}$

1) $\lim_{t \rightarrow \infty} \int_t^0 \frac{dx}{1+x^2} = \lim_{t \rightarrow -\infty} \arctan(x) \Big|_t^0 = \lim_{t \rightarrow -\infty} (\arctan(0) - \arctan(t)) = \boxed{\frac{\pi}{2}}$
 $= \boxed{\pi}$

2) $\lim_{t \rightarrow \infty} \int_0^t \frac{dx}{1+x^2} = \lim_{t \rightarrow \infty} \arctan(x) \Big|_0^t = \lim_{t \rightarrow \infty} (\arctan(t) - \arctan(0)) = \pi/2$

f) $\int_0^1 \ln(x) dx = \lim_{t \rightarrow 0^+} \int_t^1 \ln(x) dx = \lim_{t \rightarrow 0^+} \left(x \ln(x) - x \right) \Big|_t^1 =$
 $\Rightarrow \lim_{t \rightarrow 0^+} (-1 - t \ln(t) + t) = \boxed{-1}$ ∴ Converge.

9) $\int_0^{\infty} e^{-x} \cos(x) dx$
 $\left| \begin{matrix} u = \cos(x) \\ du = -\sin(x) dx \end{matrix} \right| \quad \left| \begin{matrix} dv = e^{-x} dx \\ v = -e^{-x} \end{matrix} \right| \quad \left| \begin{matrix} u = \sin(x) \\ du = \cos(x) dx \end{matrix} \right| \quad \left| \begin{matrix} dv = e^{-x} dx \\ v = -e^{-x} \end{matrix} \right|$

$\Rightarrow -\cos(x)e^{-x} - \int_0^{\infty} e^{-x} \sin(x) dx = -\sin(x)e^{-x} + \int_0^{\infty} e^{-x} \cos(x) dx$

$\Rightarrow -\sin(x)e^{-x} + I = \text{Substituto} \Rightarrow 2I = -\cos(x)e^{-x} + \sin(x)e^{-x} = \frac{e^{-x}(\sin(x) - \cos(x))}{2}$

$$\lim_{t \rightarrow \infty} \frac{e^{-x} (\sin(x) - \cos(x))}{2} \Big|_0^t = -\frac{1}{2} = \boxed{\frac{1}{2}}$$

$$d) \int_0^1 x \ln(x) dx = \frac{\ln(x)x^2}{2} - \frac{1}{2} \int_0^1 x dx = \lim_{t \rightarrow 0^+} \left(\frac{\ln(x)x^2}{2} - \frac{x^2}{4} \right) \Big|_t^1$$

$$\left| \begin{array}{l} u = \ln(x) \quad dv = x dx \\ du = \frac{1}{x} dx \quad v = \frac{x^2}{2} \end{array} \right|$$

$$= \lim_{t \rightarrow 0^+} \left(-\frac{1}{4} - \left(\frac{\ln(t)t^2}{2} - \frac{t^2}{4} \right) \right) = \boxed{-\frac{1}{4}} \quad \therefore \text{Podemos asegurar que converge.}$$

$$f) \int_1^{\infty} \frac{\sin(x)}{x^2} dx = \quad \text{Por criterio de comparación}$$

$$|f(x)| \leq g(x), \text{ donde } g(x) = \frac{1}{x^2} \Rightarrow \int_1^{\infty} \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \left. -\frac{1}{x} \right|_1^t = \lim_{t \rightarrow \infty} \left(-\frac{1}{t} + 1 \right) = \boxed{1}$$

Por criterio de comparación de integrales impropias, podemos asegurar que, si $g(x)$ converge $\Rightarrow f(x)$ también converge.

- ✓ 1. Encontrar la primitiva F de $f(x) = x + \cos(x)$ que pasa por el punto $(0, 4)$.
2. Sin realizar el cálculo de la integral, justificar las siguientes igualdades y desigualdades:

$$a) \int_{-5}^5 x^4 dx = 2 \int_0^5 x^4 dx \quad b) \int_0^4 (x-2)^3 dx = 0$$

$$c) \int_{-99}^{99} (ax^3 + bx^2 + cx) dx = 2 \int_0^{99} bx^2 dx$$

3. Trazar la región limitada por las curvas dadas y calcular su área:

- a) $y = 1/x$, $y = 1/x^2$, $x = 1$, $x = 2$
- b) $y = e^x$, $y = e^{-x}$, $x = -2$, $x = 1$
- c) $y = x + 6$, $y = x^3$, $x = -2$, $2y + x = 0$

4. Usar el cálculo integral para calcular el área de los triángulos con vértices:

- a) $(0,0)$; $(1,8)$; $(4,3)$.
- b) $(-2,5)$; $(0,-3)$; $(5,2)$.

- ✓ 5. Calcular el área de la región limitada por la parábola $y = x^2$, la tangente a ella en el punto $(1,1)$ y el eje x .

$$1) \int x - \cos(x) = \frac{x^2}{2} - \sin(x) + C \Rightarrow F(0) = C \therefore \boxed{C=4}$$

$$\text{La primitiva de la función es } \boxed{F(x) = \frac{x^2}{2} - \sin(x) + 4}$$

$$2) a) \int_{-5}^5 x^4 dx = 2 \int_0^5 x^4 dx = \text{yo subingo que, como ambas son pares obtenes el mismo}$$

$$b) \int_0^4 (x-2)^3 dx = 0$$

Podemos observar que la función es impar, por lo que en $x=2$ la función cambia de dirección y se anula completamente.

$$c) \int_{-99}^{99} (ax^3 + bx^2 + cx) dx = 2 \int_{-99}^{99} bx^2 dx$$

• Sabemos que ax^3 es impar y cx también $\Rightarrow$ Son iguales a 0

③ a) $y = \frac{1}{x}$, $y = \frac{1}{x^2}$. $x_1 = 1$, $x_2 = 2$

$\Rightarrow \frac{1}{x} = \frac{1}{x^2} \Rightarrow \boxed{x = 0}$

$\int_1^2 \left(\frac{1}{x} - x^2 \right) dx = \ln(x) + \frac{1}{x} \Big|_1^2 = \left(\ln(2) + \frac{1}{2} \right) - 1 = \boxed{\ln(2) - \frac{1}{2}}$

b) $y = e^x$, $y = e^{-x}$, $x_1 = -2$, $x_2 = 2$
 $e^x = \frac{1}{e^x}$ $x=0$ $x_1 = x_2 \Rightarrow \int_{-2}^0 (e^{-x} - e^x) dx + \int_0^2 (e^x - e^{-x}) dx$

① $= -e^x - e^x \Big|_{-2}^0 = \boxed{-2 + e^4 + e^{-2}}$
 $\Rightarrow \boxed{e + e^4 + e^{-2} + e^{-1} - 4}$

② $= e^x + e^{-x} \Big|_0^2 = \boxed{(e + e^{-1}) - 2}$

c) c) $y = x + 6$, $y = x^3$, $x = -2$, $2y + x = 0$ $x_1 = -2$, $x_2 = -\frac{x}{2}$

$x + 6 = x^3$
 $0 = x^3 - x - 6 \Rightarrow \boxed{x = 2}$

5. Calcular el área de la región limitada por la parábola $y = x^2$, la tangente a ella en el punto (1,1) y el eje x.

$f(x) = x^2$, $f'(x) = 2x = f'(1) = \boxed{2}$

la recta tangente es de la forma $y = 2(x-1) + 1 \Rightarrow \boxed{y = 2x - 1}$

$\Rightarrow x^2 - 2x + 1 = 0$ por lo que $\boxed{x = 1}$

$(x-1)^2 = 0$

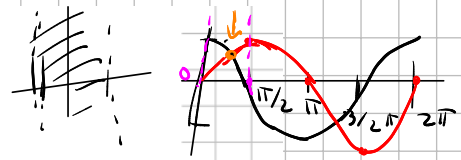
la tangente corta en $\boxed{x = \frac{1}{2}}$ $\Rightarrow \int_{\frac{1}{2}}^1 (2x-1) - x^2 dx = x^2 - x - \frac{x^3}{3} \Big|_{\frac{1}{2}}^1$
 $= -\frac{1}{3} + \frac{1}{4} - \frac{1}{2} - \frac{1}{24} = -\frac{1}{3} - \frac{7}{24} = -\frac{1}{3} + \frac{7}{24} = \frac{-8+7}{24} = \boxed{\frac{1}{24}}$

9. Trazar la región limitada por las curvas dadas y calcular su área:

a) $y = 4x^2$, $y = x^2 + 3$

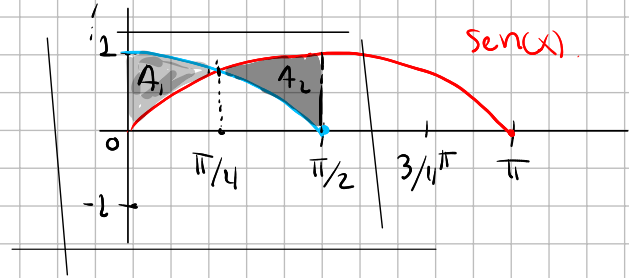
b) $y = \cos(x)$, $y = \sin(x)$, $x = 0$, $x = \pi/2$.

c) $y = |x|$, $y = (x+1)^2 - 7$, $x = -4$



b). $y = \cos(x)$, $y = \sin(x)$

$x = 0$, $x = \pi/2$



$\cos(x) = \sin(x)$

$\cos(\pi/4) = \sin(\pi/4)$

$\Rightarrow \int_0^{\pi/4} (\cos(x) - \sin(x)) dx + \int_{\pi/4}^{\pi/2} (\sin(x) - \cos(x)) dx$

① $\sin(x) + \cos(x) \Big|_0^{\pi/4} = \sqrt{2} - 1$

② $-\cos(x) - \sin(x) \Big|_{\pi/4}^{\pi/2} = -2 - \left(-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right) = \sqrt{2} - 1$

$A = 2(\sqrt{2} - 1)$